

Test ID	Category	Description	Input/Action	Expected output	Actual output	Pass/Fail
T01	File load	Load a valid csv file	Enter a valid path to data.csv	File loads without errors; displays main menu	As expected.	Pass
T02	File load	Load an invalid csv file	Enter bad_path.csv	Catches IOException and generates a new data.csv	As expected.	Pass
T03	Menu	Enter out-of-bounds menu option	Enter 9 at Main menu	"Invalid input. Selection must be within range." Prompt loops.	As expected.	Pass
T04	Menu	Enter non-integer menu option	Enter A at Main menu	"Invalid input. Must be an integer." Prompt loops.	As expected.	Pass
T05	Statistics	Calculate Total Readings	Option 1, then Option 1	Displays the exact integer count of all valid rows in CSV.	As expected.	Pass
T06	Statistics	Calculate Average	Filter by light, if none exist, select Option 2	"There are no readings for light sensors." gracefully handled.	As expected.	Pass
T07	Statistics	Unsafe Readings calculation	Filter by temperature, select Option 5	Returns count of temps < 18 or > 30 based on safe ranges.	As expected.	Pass
T08	Add data	Add reading with invalid date	Option 4, Type 1, zone 1, Val 25, Day 31, Month 2	"february can only have 28 days". Prompt loops.	As expected.	Pass
T09	Add data	Add valid reading	Option 4, type 1, zone 1, val 25, date 12/3/2026 14:30	Data appended to array and written to CSV file.	As expected.	Pass
T10	Delete data	Delete reading with bad ID format	Option 5, ID 12345	"Invalid sensor format: 12345". Prompt loops.	As expected.	Pass
T11	Delete data	Delete valid reading	Option 5, TMP332	Success message. Array shrinks. CSV is rewritten without record.	As expected.	Pass